

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Display
 Manufacturer: Samsung
 Firmware Version: N/A
 Model(s): UN43TU8000, UN50TU8000, UN55TU8000, UN65TU8000, UN75TU8000, UN85TU8000

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.17.0000-b002	2.20.0.16

Version History

Module Version	Date	Notes
1_0_1_0	1/30/2024	Initial Version

Module Notes

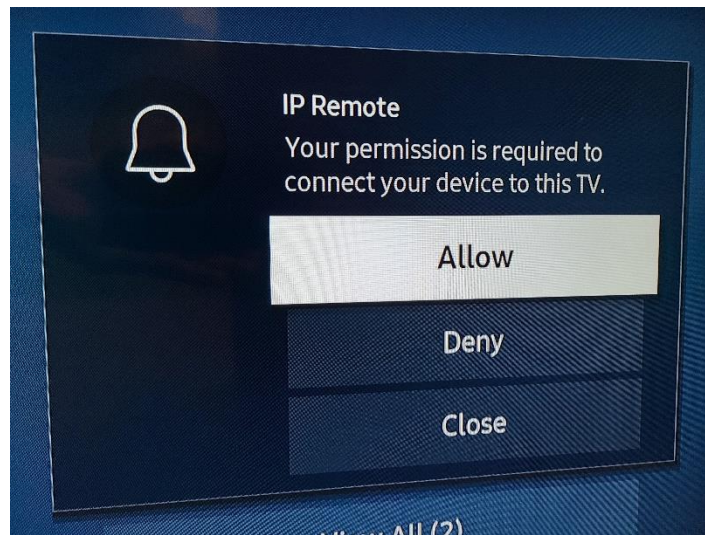
- SSLVerifyMode variable must be set accordingly. Default value is 'On'.
- Status is not supported because the LAN port closes upon powering down the display. To power on the display, please use either the IR remote or Wake-on-LAN.

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='UN43TU8000')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='UN43TU8000')
HTTPClass
InterfaceName = ModuleName.HTTPClass('192.168.254.254', 1516, Model='UN43TU8000', SSLVerifyMode='On')

Refresh Token Command

- This device requires a token to be given to the client for Ethernet control to work.
- After the module is programmed, use the 'RefreshToken' command.
- After this command is used, an 'IP Remote' prompt should appear.
- Refer to the image below for reference:



- Use the IR remote or driver to 'Allow' the connection.
- The module will be able to control the display now.

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AspectRatio	'4:3'	'16:9'	'Custom'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command	Value		
AudioMute	None		
# AudioMute example InterfaceName.Set('AudioMute', None)			
Command	Value	Value	
Channel	'Up'	'Down'	
# Channel example InterfaceName.Set('Channel', 'Up')			
Command	Value	Value	Value
Input	'TV'	'AV'	'HDMI 1'
	'HDMI 2'	'HDMI 3'	
# Input example InterfaceName.Set('Input', 'TV')			
Command	Value		
Keypad	'0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command	Value	Value	Value
MenuNavigation	'Up'	'Down'	'Left'
	'Right'	'Menu'	'Enter'
	'Return'	'Exit'	
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command	Value	Value	
Power	'On'	'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command	Value		
Volume	0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value '4:3'	Value '16:9'	
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command Channel	Value 'Up'	Value 'Down'	
# Channel example InterfaceName.Set('Channel', 'Up')			
Command Input	Value 'TV' 'HDMI 2'	Value 'AV' 'HDMI 3'	Value 'HDMI 1'
# Input example InterfaceName.Set('Input', 'TV')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command MenuNavigation	Value 'Up' 'Right' 'Return'	Value 'Down' 'Menu' 'Exit'	Value 'Left' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command PowerOff	Value None		
# PowerOff example InterfaceName.Set('PowerOff', None)			
Command RefreshToken ¹	Value None		
# RefreshToken example InterfaceName.Set('RefreshToken', None)			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

¹ Please refer to 'Refresh Token Command' section above for details.

Cable and Adapter Requirements

Captive Screw to 3.5mm TRS RS-232 Serial Cable. Note that a Samsung-specific Ex-Link to USB adapter is also necessary. Please contact the manufacturer for more information.

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

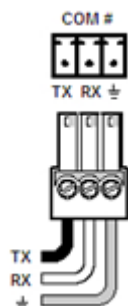
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Please contact the manufacturer for pin diagram information.



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (HTTPS)

Default Port: 1516

Logon Credentials No

Supported:

Multi-Connection Undetermined

Capabilities:

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Please ensure the following setting is configured for the module to work. Using the IR remote, open the Menu of the display:

- General → Network → Expert Settings → IP Remote → Enable

Appendix A. Set Commands(Serial)

Aspect Ratio 16:9	\x08"\x0B\x0A\x01\x00\xC0
Aspect Ratio 4:3	\x08"\x0B\x0A\x01\x04\xBC
Aspect Ratio Custom	\x08"\x0B\x0A\x01\x0B\xB5
Audio Mute None	\x08"\x02\x00\x00\x00\xD4
Audio Mute Off	\x08"\x02\x00\x00\x00\xD4
Audio Mute On	\x08"\x02\x00\x00\x00\xD4
Channel Down	\x08"\x03\x00\x02\x00\xD1
Channel Up	\x08"\x03\x00\x01\x00\xD2
Input AV	\x08"\x0A\x00\x01\x00\xCB
Input HDMI 1	\x08"\x0A\x00\x05\x00\xC7
Input HDMI 2	\x08"\x0A\x00\x05\x01\xC6
Input HDMI 3	\x08"\x0A\x00\x05\x02\xC5
Input TV	\x08"\x0A\x00\x00\x00\xCC
Keypad 0	\x08"\x0D\x00\x00\x11\xB8
Keypad 9	\x08"\x0D\x00\x00\x0E\xBB
Menu Navigation Down	\x08"\x0D\x00\x00ah
Menu Navigation Enter	\x08"\x0D\x00\x00. \x9B
Menu Navigation Exit	\x08"\x0D\x00\x00- \x9C
Menu Navigation Left	\x08"\x0D\x00\x00ed
Menu Navigation Menu	\x08"\x0D\x00\x00\x1A\xAF
Menu Navigation Return	\x08"\x0D\x00\x00Xq
Menu Navigation Right	\x08"\x0D\x00\x00bg
Menu Navigation Up	\x08"\x0D\x00\x00` i
Power Off	\x08"\x00\x00\x00\x01\xD5
Power On	\x08"\x00\x00\x00\x02\xD4
Volume 0	\x08"\x01\x00\x00\x00\xD5
Volume 100	\x08"\x01\x00\x00dq